

Artemy Kolchinsky

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EDUCATION	Indiana University (Bloomington, IN, USA), 2015 Ph.D. in Informatics (focus in Complex Systems), Minor in Cognitive Science Thesis: “Measuring Scales: Integration and Modularity in Complex Systems” Committee: Luis M. Rocha (chair), Yong-Yeol Ahn, Randall Beer, Alessandro Flammini, Olaf Sporns New York University (New York, NY, USA), 2004 B.A. Magna Cum Laude, Individualized Study (concentration in Complex Systems)
ACADEMIC POSITIONS	6/2023-Present Universitat Pompeu Fabra (Barcelona, Spain) Marie Curie postdoctoral fellow (host: Ricard Solé) 1/2022-5/2023 University of Tokyo (Tokyo, Japan) Project researcher at the Universal Biology Institute (host: Sosuke Ito) 12/2015-7/2021 Santa Fe Institute (Santa Fe, NM, USA) Program postdoctoral fellow (advisor: David H. Wolpert)
INDUSTRY	Summer 2014 LinkedIn Corporation (Mountain View, CA, USA) Data science internship
PUBLICATIONS	A. Kolchinsky , “Thermodynamics of Darwinian evolution in molecular replicators”, <i>Philosophical Transactions of the Royal Society B</i> , 2025. pdf link K. R. Thomsen, A. Kolchinsky , S. Rasmussen, “Protocellular energy transduction, information, and fitness”, <i>Philosophical Transactions of the Royal Society B</i> , 2025. pdf link S. Bartlett, A. W. Eckford, M. Egbert, M. Lingam, A. Kolchinsky , A. Frank, G. Ghoshal, “The Physics of Life: Exploring Information as a Distinctive Feature of Living Systems”, <i>PRX Life</i> , 2025. pdf link R. Nagayama, K. Yoshimura, A. Kolchinsky , S. Ito, “Geometric thermodynamics of reaction-diffusion systems: Thermodynamic trade-off relations and optimal transport for pattern formation”, <i>Physical Review Research</i> , 2025. pdf link A. Kolchinsky , I. Marvian, C. Gokler, Z. Liu, P. Shor, O. Shtanko, K. Thompson, D. Wolpert, S. Lloyd, “Maximizing free energy gain”, <i>Entropy</i> , 2025. pdf link A. Kolchinsky , “Thermodynamic dissipation does not bound replicator growth and decay rates”, <i>The Journal of Chemical Physics</i> , 2024. pdf link (2024 JCP Emerging Investigators Special Collection) R. Solé, C. P. Kempes, B. Corominas-Murtra, M. De Domenico, A. Kolchinsky , M. Lachmann, E. Libby, S. Saavedra, E., D. Wolpert, “Fundamental constraints to the logic of living systems”, <i>Royal Society Interface</i> , 2024. pdf link P.M. Riechers, C. Gupta, A. Kolchinsky , M. Gu, “Thermodynamically ideal quantum state inputs to any device”, <i>PRX Quantum</i> , 2024. pdf link A. Kolchinsky , “Partial information decomposition: redundancy as information bottleneck”, <i>Entropy</i> , 2024. pdf link J. Piñero, R. Solé, A. Kolchinsky , “Optimization of nonequilibrium free energy harvesting illustrated on bacteriorhodopsin”, <i>Physical Review Research</i> , 2024. pdf link A. Kolchinsky , N. Ohga, S. Ito, “Thermodynamic bound on spectral perturbations, with applications to oscillations and relaxation dynamics”, <i>Physical Review Research</i> , 2024. pdf link D.R. Sowinski, J. Carroll-Nellenback, R.N. Markwick, J. Piñero, M. Gleiser, A. Kolchinsky , G. Ghoshal, A. Frank, “Semantic Information in a model of resource gathering agents”, <i>PRX Life</i> , 2023. pdf link A. Kolchinsky , “Generalized Zurek’s bound on the cost of an individual classical or quantum computation”, <i>Physical Review E</i> , 2023. pdf link

M. Aguilera and **A. Kolchinsky**, “Quantifying higher-order entropy production in organized nonequilibrium states”, *Proceedings of the ALIFE 2023*, 2023. [pdf link](#)

N. Ohga, S. Ito, **A. Kolchinsky**, “Thermodynamic bound on the asymmetry of cross-correlations”, *Physical Review Letters*, 2023. [pdf link](#) (Editors’ Suggestion; Featured in *Physics*)

K. Yoshimura, **A. Kolchinsky**, A. Dechant, S. Ito, “Housekeeping and excess entropy production for general nonlinear dynamics”, *Physical Review Research*, 2023. [pdf link](#)

F.C. Sheldon, **A. Kolchinsky**, F. Caravelli, “Computational capacity of LRC, memristive, and hybrid reservoirs”, *Physical Review E*, 2022. [pdf link](#)

A. Kolchinsky, “A novel approach to the partial information decomposition”, *Entropy*, 2022. [pdf code link](#)

A. Kolchinsky and D.H. Wolpert, “Dependence of integrated, instantaneous, and fluctuating entropy production on the initial state in quantum and classical processes”, *Physical Review E*, 2021. [pdf link](#)

A. Kolchinsky and D.H. Wolpert, “Work, entropy production, and thermodynamics of information under protocol constraints”, *Physical Review X*, 2021. [pdf link](#)

A. Kolchinsky and D.H. Wolpert, “Entropy production given constraints on the energy functions”, *Physical Review E*, 2021. [pdf link](#)

A. Kolchinsky and D.H. Wolpert, “Thermodynamic costs of Turing machines”, *Physical Review Research*, 2020. [pdf link](#)

D.H. Wolpert and **A. Kolchinsky**, “The thermodynamics of computing with circuits”, *New Journal of Physics*, 2020. [pdf link](#)

A. Kolchinsky and B. Corominas-Murtra, “Decomposing information into copying versus transformation”, *Royal Society Interface*, 2020. [pdf link](#)

A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *Journal of Statistical Mechanics*, 2019. [pdf code link](#)

A. Kolchinsky, B.D. Tracey, D.H. Wolpert, “Nonlinear information bottleneck”, *Entropy*, 2019. (*Entropy* 2021 Best Paper Award) [pdf link](#)

A. Berdahl, C. Brelsford, C. De Bacco, M. Dumas, V. Ferdinand, J.A. Grochow, L. Hébert-Dufresne, Y. Kallus, C.P. Kempes, **A. Kolchinsky**, D. B. Larremore, E. Libby, E.A. Power, C.A. Stern, B.D. Tracey, “Dynamics of beneficial epidemics”, *Scientific Reports*, 2019. [pdf link](#)

E.A. Hobson, V. Ferdinand, **A. Kolchinsky**, J. Garland, “Rethinking animal social complexity measures with the help of complex systems concepts”, *Animal Behaviour*, 2019. [pdf link](#)

A. Kolchinsky, B.D. Tracey, S. Van Kuyk, “Caveats for information bottleneck in deterministic scenarios”, *ICLR*, 2019. [pdf code link](#)

D.H. Wolpert, **A. Kolchinsky**, J.A. Owen, “A space–time tradeoff for implementing a function with master equation dynamics”, *Nature Communications*, 2019. [pdf link](#)

A. Avena-Koenigsberger, X. Yan, **A. Kolchinsky**, M. van den Heuvel, P. Hagmann, O. Sporns, “A spectrum of routing strategies for brain networks”, *PLoS Computational Biology*, 2019. [pdf link](#)

J.A. Owen, **A. Kolchinsky**, D.H. Wolpert, “Number of hidden states needed to physically implement a given conditional distribution”, *New Journal of Physics*, 2019. [pdf link](#) (correction)

A. Kolchinsky and D.H. Wolpert, “Semantic information, autonomous agency, and nonequilibrium statistical physics”, *Royal Society Interface Focus*, 2018. [pdf code link](#)

A.M. Saxe, Y. Bansal, J. Dapello, M. Advani, **A. Kolchinsky**, B.D. Tracey, D.D. Cox, “On the information bottleneck theory of deep learning”, *ICLR*, 2018. [pdf code link](#)

A. Kolchinsky, N. Dhande, K. Park, Y.Y. Ahn, “The Minor Fall, the Major Lift: Inferring emotional valence of musical chords through lyrics”, *Royal Society Open Science*, 2017. [pdf data code link](#)

A. Kolchinsky, D.H. Wolpert, “Dependence of dissipation on the initial distribution over states”, *Journal of Statistical Mechanics*, 2017. [pdf link](#)

A. Kolchinsky, B.D. Tracey, “Estimating mixture entropy with pairwise distances”, *Entropy*, 2017. (correction) [pdf code link](#)

A. Kolchinsky, A.J. Gates, L.M. Rocha, “Modularity and the spread of perturbations in complex dynamical

systems,” *Physical Review E*, 2015. [pdf code link](#)

A. Kolchinsky, A. Lourenço, H. Wu, L. Li, L.M. Rocha, “Extraction of pharmacokinetic evidence of drug-drug interactions from the literature,” *PLOS One*, 2015. [pdf link](#)

A. Kolchinsky, M.P. van den Heuvel, A. Griffo, P. Hagmann, L.M. Rocha, O. Sporns, J. Goñi, “Multi-scale integration and predictability in resting state brain activity,” *Frontiers in Neuroinformatics*, 2014. [pdf link](#)

A. Rossi, F.J. Parada, **A. Kolchinsky**, A. Puce, “Neural correlates of apparent motion perception of impoverished facial stimuli I: A comparison of ERP and ERSP activity,” *NeuroImage*, 2014. [pdf link](#)

A. Kolchinsky, A. Lourenço, L. Li, L.M. Rocha, “Evaluation of linear classifiers on articles containing pharmacokinetic evidence of drug-drug interactions,” *Proc Pacific Symposium on Biocomputing*, 2013. [pdf link](#)

A. Kolchinsky and L.M. Rocha, “Prediction and modularity in dynamical systems,” *Proc of European Conf. on the Synthesis and Simulation of Living Systems (ECAL)*, 2011. [pdf link](#)

A. Kolchinsky, A. Abi-Haidar, J. Kaur, A.A. Hamed, L.M. Rocha, “Classification of protein-protein interaction full-text documents using text and citation network features,” *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, 2010. [pdf link](#)

PREPRINTS

A. Kolchinsky, “Comment on ‘Dissipation bounds the coherence of stochastic limit cycles’”, arXiv:2510.14101, 2025. [arxiv](#)

J. Bauermann, G. Bartolucci, **A. Kolchinsky**, “Spatiotemporal organization of chemical oscillators via phase separation”, arXiv:2507.16030, 2025. [arxiv](#)

G.-H. Xu, **A. Kolchinsky**, J.-C. Delvenne, and S. Ito, “Thermodynamic geometric constraint on the spectrum of Markov rate matrices”, arXiv:2507.08938, 2025. [arxiv](#)

J. Piñero, **A. Kolchinsky**, S. Redner, R. Solé, “Neutral theory of cooperators”, arXiv:2506.09737, 2025. [arxiv](#)

M. Aguilera, S. Ito, **A. Kolchinsky**, “Inferring entropy production in many-body systems using nonequilibrium MaxEnt”, arXiv:2505.10444, 2025. [arxiv](#)

J. Piñero, D. R. Sowinski, G. Ghoshal, A. Frank, **A. Kolchinsky**, “Information bounds production in replicator systems”, arXiv:2501.00396, 2024. [arxiv](#)

A. Kolchinsky, A. Dechant, K. Yoshimura, S. Ito, “Generalized free energy and excess entropy production for active systems”, arXiv:2412.08432, 2024. [arxiv](#)

A. Kolchinsky, A. Dechant, K. Yoshimura, S. Ito, “Information geometry of excess and housekeeping entropy production”, arXiv:2206.14599, 2022. [arxiv](#)

C. Gokler, **A. Kolchinsky**, Z. Liu, I. Marvian, P. Shor, O. Shtanko, K. Thompson, D. Wolpert, S. Lloyd, “When is a bit worth much more than $kT \ln 2$?”, arXiv:1705.09598, 2017. [arxiv](#)

INVITED TALKS

11/2025 *Kyoto Workshop on Quantum Thermodynamics & Stochastic Thermodynamics*, Kyoto University, Japan

10/2025 *Autocatalysis in Reaction Networks Seminar (ARN)* (online)

9/2025 *ILIAD2: ODYSSEY, conference on mathematical alignment*, Berkeley, CA, USA

6/2025 *CRC Colloquium*, Free University of Berlin, Berlin, Germany

5/2025 *Joint European Thermodynamics Conference*, Belgrade, Serbia

11/2024 *Universal Biology Institute Seminar Series*, University of Tokyo, Japan

9/2024 *Information in Matter Colloquium*, AMOLF, Amsterdam, Netherlands

7/2024 *Physics Department Seminar*, University of Barcelona, Barcelona, Spain

2/2024 *Seminar*, Barcelona Collaboratorium for Modelling and Predictive Biology, Barcelona, Spain

11/2023 *Mini-workshop: Non-equilibrium thermodynamics and biology*, Biofisika Institute, Bilbao, Spain

11/2023 *Seminar*, Basque Center for Applied Mathematics, Bilbao, Spain

7/2023 *Information Engines at the Frontiers of Nanoscale Thermodynamics*, Telluride, CO, USA

6/2023 *Webinar: Mathematics, Physics & Machine Learning*, Instituto Superior Técnico, Lisbon, Portugal

6/2023 *Conference: Decomposing Multivariate Information in Complex Systems*

Max Planck Institute for the Physics of Complex Systems, Dresden, Germany

9/2022 *Seminar*, Dutch Institute of Emergent Phenomena, University of Amsterdam, Netherlands

8/2022 *Seminar*, Earth-Life Science Institute (ELSI), Tokyo, Japan
6/2022 *Evolution of complexity from the statistical physics perspective*, Yerevan Physics Institute, Armenia
6/2022 *Yagami Statistical Physics Seminar*, Keio University, Japan
5/2022 *Workshop on Stochastic Thermodynamics III*, University of Tokyo, Japan
4/2022 *Cross Labs Workshop: Is AI Extending the Mind?*, Cross Labs, Japan
2/2022 *Physics Department Colloquium Series*, University of Rochester, NY, USA
12/2021 *Universal Biology Institute Seminar Series*, University of Tokyo, Japan
02/2021 *Workshop: Origins of Life: The Possible and the Actual*, Santa Fe Institute, NM, USA
7/2020 *ICTP Seminar Series*, Abdus Salam International Center for Theoretical Physics, Trieste, Italy
2/2020 *AI Seminar Series*, Information Sciences Institute, Los Angeles, CA, USA
7/2019 *ISTI Seminar Series*, Los Alamos National Lab, Los Alamos, NM, USA
6/2018 *Connectomics Lecture Series*, Universidad Diego Portales, Santiago, Chile
4/2018 *Meeting of the Society for the Neural Control of Movement*, Santa Fe, NM, USA
11/2017 Seoul National University, South Korea
8/2017 *Workshop: Thermodynamics & Computation: Towards a New Synthesis*, Santa Fe Institute, NM, USA
10/2016 *Statistical Physics, Information Processing and Biology*, Santa Fe Institute, NM, USA
2/2016 Information Sciences Institute, Los Angeles, CA, USA

AWARDS & FELLOWSHIPS

2022 Marie Curie Individual Fellowship (HORIZON-MSCA-2021-PF-01, #101068029)
“NETOLIFE: Nonequilibrium thermodynamics of the origin of life”

GRANTS

2023 John Templeton Foundation (#62828), \$947,926, Co-PI
“Goal-directed behavior and the origin of life”
2022 John Templeton Foundation (#62417), \$627,227, Co-PI
“Information architectures that enable life: the emergence of meaning”
2019 Foundational Questions Institute (FQXi-RFP-IPW-1912), \$118,100, Co-Investigator (PI: David Wolpert)
“The role of constraints in the thermodynamics of intelligence”
2016 Foundational Questions Institute (FQXi-RFP-1622), \$128,319, Co-Investigator (PI: David Wolpert)
“Observers as self-maintaining non-equilibrium systems”
2016 NSF INSPIRE (CHE-1648973), \$999,947, co-author (PI: David Wolpert)
“Tradeoffs in the thermodynamics of computation: a new paradigm for biological information-processing”
2016 NSF (1620462), \$770,000, co-author (PI: David Wolpert)
“Information Networks and the Evolution of Social Organizations”

POPULAR SCIENCE & OUTREACH

Articles

A. Kolchinsky, “The cost of sending a bit across a living cell”, *Physics Magazine*, 2023. [link](#)

Public talks

4/2018 *SITE Santa Fe* (contemporary art museum), Santa Fe, NM, USA
Public talk on “Life, entropy, and the 2nd law of thermodynamics”

TEACHING

Online courses

2018 “Fundamentals of machine learning” (w/ B.D. Tracey), Santa Fe Institute *Complexity Explorer*
Designed and delivered a 12-part online course on basics of machine learning ([link](#))

Workshops, lectures, and tutorials

6/2019 Santa Fe Institute Complex Systems Summer School, NM, USA
Two-part lecture series on machine learning and its research frontiers
3/2019 Santa Fe Institute, NM, USA
Tutorial on “Machine learning with TensorFlow”

6/2017, Santa Fe Institute, NM, USA
 6/2018 Tutorial on introduction to programming and data analysis in Python (w/ V. Ferdinand)
 11/2017 Seoul National University, Seoul, South Korea
 Week-long workshop organized w/ V. Ferdinand
 “Thermodynamics, evolution, and inference through the lens of information theory”
 11/2017 ACTioN/Trustee Meeting, Santa Fe Institute, NM, USA
 Lecture on “Machine learning: A guide for the perplexed” (w/ B. D. Tracey)
 5/2010 Instituto Gulbenkian de Ciência, Oeiras, Portugal
 Assisted with week-long “Bayesian brain” educational module

Teaching Assistant at Indiana University, Bloomington, IN

Spring 2014 “I400 Large-scale Social Phenomena” with Simon DeDeo ([link](#))
 Spring 2011 “I201 Math and logic foundations of Informatics” with Erik Wennstrom
 Fall 2010 “I485 Biologically Inspired Computing” ([link](#)) with Luis M. Rocha
 Developed and implemented a 5-part problem set for a computational lab
 Fall 2008- “I210 Information Infrastructure” (various instructors)
 Spring 2009 Ran lab to assist with Python programming course

ADVISING

PhD dissertation committees

2025 *André Gomes*, PhD, Electrical & Computer Engineering, Instituto Superior Técnico, Lisbon, Portugal
 Thesis: “Advances in Partial Information Decomposition”

Undergraduate projects supervised

2018 *Nicolas Freitas*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA
 Project: “Scaling of information in biochemical systems”
 2017 *Francis Cavanna*, Santa Fe Institute Summer REU Program, Santa Fe, NM, USA
 Project: “Investigating the relationship between criticality and Landauer costs using the Ising model”

ACADEMIC SERVICE

Reviewer

General: *PNAS*, *Nature Communications*, *npj Complexity*, *Interface Focus*
 Physics: *PRL*, *PRX*, *PRX Quantum*, *PRR*, *PRE*, *NJP*, *Frontiers in Physics*, *J of Physics A*, *Physica A*, *EPL*
 Biology: *Proc R Soc B*, *Phil Trans B*, *PLoS Comp Bio*, *Theory in the Biosciences*, *BioSystems*
 Information theory and machine learning: *Entropy*, *ICLR*, *IEEE TPAMI*, *Kybernetika*, *Applied Sciences*

Guest editor

Entropy special issue on “Thermodynamics and Information Theory of Living Systems” [link](#)

Grant Review Panels

2025 Interdisciplinary Assessment Board for Artificial Intelligence, Spanish Research Agency, reviewer
 2021- John Templeton Foundation, external reviewer
 2023 Swiss National Science Foundation, external reviewer
 2020 European Research Council (ERC), Starting Grant Panel, external reviewer

Conference Organization

12/2025 Kyoto Workshop on Quantum Thermodynamics and Stochastic Thermodynamics, co-organizer
 Kyoto University, Japan ([link](#))
 7/2024 Information-Driven States of Matter, co-organizer
 University of Rochester, NY, USA ([program](#)) [press](#)

Seminar Organization

2023- Co-organized ongoing “Complex Systems Lab” seminar series
[Barcelona Collaboratorium for Modelling and Predictive Biology](#), Barcelona, Spain
 2008- Organized a weekly discussion group on complexity and cognitive science ([link](#))
 2013 Indiana University, Bloomington, IN, USA

LANGUAGES Fluent: English, Russian, Spanish. Basic: Portuguese